

Candidate Name : _____

CT Group : _____

Index no : _____

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examination

MATHEMATICS
Higher 2



9740/1

Tuesday

9th September 2008

Additional materials: Answer Papers
List MF15

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your full name, index number and CT group on all the work you hand in.

Write in dark blue or black pen on both sides of the papers.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, arrange your solutions in numerical order and fasten them securely together with the question paper on the top.

For Examiner's Use					
Qn	Marks	Qn	Marks	Qn	Marks
1		5		9	
2		6		10	
3		7		11	
4		8		12	
				Total	

This question paper consists of 4 printed pages.

1. A cubic curve passes through the points $\left(-1, \frac{1}{2}\right)$, $(2, 14)$ and $(4, 123)$. Given that the gradient of the curve at the point where $x=1$ is $\frac{9}{2}$, find the corresponding y -coordinate at this point. [4]
2. (a) Find all the exact roots of the equation $z^4 = 2 - 2\sqrt{3}i$, leaving your answers in the form $re^{i\theta}$. [3]
- (b) Solve the simultaneous equations
- $$4iz + 3w = 7 + 5i,$$
- $$2z - (1+i)w = 10 - i,$$
- giving z and w in the form $a + ib$, where a and b are real. [3]
3. Prove by induction that, for all positive integers n , $\sum_{r=1}^n r(2^r) = 2^{n+1}(n-1) + 2$. [4]
- Hence find $\sum_{r=1}^n r(2^{r+1} - 2)$ in terms of n . [3]
4. A sequence $u_1, u_2, u_3, \dots$ is such that $u_1 = \frac{e}{30}$ and
- $$u_n - u_{n-1} = \frac{e}{4^{2n-1}}, \text{ for all } n \geq 2.$$
- Use the method of difference to show that $u_N = \frac{e}{20} \left[1 - \frac{1}{3} \left(\frac{1}{16} \right)^{N-1} \right]$ for all $N \geq 1$. [4]
- Hence or otherwise, determine if
- (i) u_n is a convergent or divergent sequence, [2]
- (ii) $\sum_{n=1}^N u_n$ is a convergent or divergent series. [2]
5. Express $f(x) = \frac{x-5}{(x+1)(3x+2)}$ in partial fractions and hence, or otherwise, obtain $f(x)$ as a series expansion in ascending powers of x as far as the term in x^3 . State the range of values of x for which the expansion is valid. Find also, the coefficient of x^n in this expansion, where $n > 1$. [8]

[Turn over]

6. An arithmetic progression has 889 terms. The sum of all the even-numbered terms of the progression is 408480. The 1st term, 9th term, and the 21st term of the progression are three consecutive terms of a geometric progression. Find the first term and the common difference of the arithmetic progression. [8]
7. It is given that $\ln(1+y) = \tan^{-1} x$.
- (i) Prove that $(1+x^2)\frac{dy}{dx} = 1+y$. [2]
- (ii) Find the first three non-zero terms in the Maclaurin's series for y . [3]
- (iii) Hence, find the approximate value for $\int_0^{\frac{1}{2}} (e^{\tan^{-1} x} - 1) dx$. Deduce the percentage error with the aid of a Graphic Calculator. [4]
8. The quantities x and y are related by the differential equation $x\frac{dy}{dx} = y(1-x-y)$.
- (i) Show that this differential equation may be reduced by means of the substitution $zy = x$ to $\frac{dz}{dx} = z+1$. Hence, find the general solution of this differential equation, expressing y in terms of x . [5]
- (ii) Sketch the solution curve which passes through the point $(-2,1)$, stating clearly the equations of the asymptotes. [4]
9. Sketch the curve $y = k + x \cos(x^2)$ and the line $y = k$, where $0 \leq x \leq \sqrt{\pi}$ and k is a positive constant. [3]
- State the transformation which would transform the curve to the graph of $y = x \cos(x^2)$. [1]
- Hence
- (i) by considering the derivative of $\sin(x^2)$, find the exact area of the region R bounded by the curve $y = k + x \cos(x^2)$ and the lines $x = \sqrt{\pi}$ and $y = k$, [3]
- (ii) calculate the volume of the solid of revolution formed when the region R is rotated completely about the line $y = k$. [2]

[Turn over]

10. The line l_1 and plane Π_1 have vector equations

$$\mathbf{r} = \mathbf{i} + 5\mathbf{j} - 4\mathbf{k} + \lambda(3\mathbf{i} + 2\mathbf{j} + 7\mathbf{k}), \quad \lambda \in \mathbb{R}$$

$$\mathbf{r} = \mathbf{i} + 5\mathbf{j} - 4\mathbf{k} + s(2\mathbf{i} + \mathbf{j} + 4\mathbf{k}) + t(3\mathbf{i} + 2\mathbf{j} + 7\mathbf{k}), \quad s, t \in \mathbb{R}$$

respectively.

- (i) State a point of intersection between l_1 and Π_1 , and hence give a geometrical interpretation about l_1 and Π_1 . [2]
- (ii) The plane Π_2 passes through the point A with coordinates $(3, 6, 1)$ and contains the line l_1 . Find, in the form $\mathbf{r} \cdot \mathbf{n} = p$, the equation of Π_2 . [2]
- (iii) The plane Π_3 has equation $x + 2y - z = 0$. Determine, with a reason, if the planes Π_1 , Π_2 and Π_3 have a common point of intersection. [3]
- (iv) Find the perpendicular distance between Π_1 and Π_3 , giving your answer in the surd form. [2]

11. A curve has parametric equations $x = \sec\left(\frac{1}{2}\theta\right)$, $y = \ln \sec\left(\frac{1}{2}\theta\right)$, where $-\pi < \theta < \pi$.

Show that, at the point with parameter θ , the gradient of the curve is $\cos\left(\frac{1}{2}\theta\right)$. [3]

- (a) Find the equation of the normal to the curve at $\theta = \frac{\pi}{3}$. [3]
- (b) Determine the rate of change of x if the gradient of the curve at $\theta = \frac{\pi}{2}$ is decreasing at a rate of 0.4 units per second. [5]

12. The curve C has equation

$$y = \frac{x^2}{2 + \lambda x}, \quad \text{where } \lambda \text{ is a positive constant.}$$

- (i) State the equations of the asymptotes of C . [3]
- (ii) Show that C has exactly two points at which $\frac{dy}{dx} = 0$. [2]
- (iii) Sketch the curve C , stating clearly the asymptotes and axes intercepts. [3]
- (iv) Hence solve the inequality $\frac{(1 + \lambda^2)x^2 + 2\lambda x}{2 + \lambda x} \geq 2$. [4]

Solution

1. Let equation of the curve be $y = ax^3 + bx^2 + cx + d$

Therefore, $\frac{dy}{dx} = 3ax^2 + 2bx + c$

Using the coordinates and gradient of curve given,

$$-a + b - c + d = \frac{1}{2} \quad \dots (1)$$

$$8a + 4b + 2c + d = 14 \quad \dots (2)$$

$$64a + 16b + 4c + d = 123 \quad \dots (3)$$

$$3a + 2b + c = \frac{9}{2} \quad \dots (4)$$

$$\begin{pmatrix} -1 & 1 & -1 & 1 \\ 8 & 4 & 2 & 1 \\ 64 & 16 & 4 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 14 \\ 123 \\ \frac{9}{2} \end{pmatrix}$$

Using GC, $a = 2$, $b = 0$, $c = -\frac{3}{2}$, $d = 1$

Therefore, equation of curve: $y = 2x^3 - \frac{3}{2}x + 1$

Thus, at $x = 1$, $y = \frac{3}{2}$.

2. (a) $z^4 = 2 - 2\sqrt{3}i = 4e^{i\left(-\frac{\pi}{3} + 2n\pi\right)}$
 $z = 4^{\frac{1}{4}} e^{i\left(-\frac{\pi}{12} + \frac{n\pi}{2}\right)}, n = 0, 1, 2, 3$
 $z = 4^{\frac{1}{4}} e^{i\left(-\frac{\pi}{12}\right)}, 4^{\frac{1}{4}} e^{i\left(\frac{5\pi}{12}\right)}, 4^{\frac{1}{4}} e^{i\left(\frac{11\pi}{12}\right)}, 4^{\frac{1}{4}} e^{i\left(\frac{17\pi}{12}\right)}$

(b) $4iz + 3w = 7 + 5i$, --- (1)

$2z - (1+i)w = 10 - i$, --- (2)

(2) $\times 2i$,

$4iz - 2i(1+i)w = (10-i)2i$

$4iz + 2w - 2iw = 20i + 2$ --- (3)

(1) - (3),

$w + 2iw = 5 - 15i \Rightarrow w = \frac{5 - 15i}{1 + 2i} = -5 - 5i$

Sub into (2),

$2z - (1+i)(-5-5i) = 10 - i$

$2z + 10i = 10 - i$

$\Rightarrow z = 5 - \frac{11}{2}i$

3. Let P_n be the statement $\sum_{r=1}^n (r(2^r)) = (2)^{n+1}(n-1) + 2, n \in \mathbb{N}, n \geq 1$.

When $n = 1$,

$$LHS = 1(2^1) = 2$$

$$RHS = (2)^2(1-1) + 2 = 2 = LHS$$

Therefore, P_1 is true.

Assume P_k is true for some $k \in \mathbb{N}, k \geq 1$, i.e. $\sum_{r=1}^k (r(2^r)) = (2)^{k+1}(k-1) + 2$.

To prove P_{k+1} also true, i.e. $\sum_{r=1}^{k+1} (r(2^r)) = (2)^{k+2}(k) + 2$:

$$\begin{aligned} LHS &= \sum_{r=1}^{k+1} (r(2^r)) \\ &= \sum_{r=1}^k (r(2^r)) + (k+1)(2^{k+1}) \\ &= (2)^{k+1}(k-1) + 2 + (k+1)(2^{k+1}) \\ &= (2)^{k+1}[(k-1) + (k+1)] + 2 \\ &= (2)^{k+1}[2k] + 2 \\ &= (2)^{k+2}(k) + 2 = RHS \end{aligned}$$

Since P_1 is true, $P_k \Rightarrow P_{k+1}$ is also true, hence by mathematical induction, P_n is true for all $n \in \mathbb{N}, n \geq 1$.

$$\begin{aligned} \sum_{r=1}^n (r(2^{r+1} - 2)) &= \sum_{r=1}^n (r(2^{r+1})) - \sum_{r=1}^n (2r) \\ &= 2 \sum_{r=1}^n (r(2^r)) - (2 + 4 + 6 + 8 + \dots + 2n) \\ &= 2 \left[(2)^{n+1}(n-1) + 2 \right] - \frac{n}{2} [2(2) + (n-1)(2)] \\ &= 2 \left[(2)^{n+1}(n-1) + 2 \right] - n(n+1) \end{aligned}$$

$$\begin{aligned}
4. \quad u_n - u_{n-1} &= \frac{e}{4^{2n-1}} \Rightarrow \sum_{n=2}^N (u_n - u_{n-1}) = \sum_{n=2}^N \left(\frac{e}{4^{2n-1}} \right) \\
&\Rightarrow \begin{cases} u_2 - u_1 \\ u_3 - u_2 \\ u_4 - u_3 \\ \vdots \\ u_{N-2} - u_{N-3} \\ u_{N-1} - u_{N-2} \\ u_N - u_{N-1} \end{cases} = e \sum_{n=2}^N \left(\frac{1}{4^{2n-1}} \right) \\
&\Rightarrow u_N - u_1 = e \left(\frac{1}{4^3} + \frac{1}{4^5} + \frac{1}{4^7} + \dots + \frac{1}{4^{2N-1}} \right) \\
&\Rightarrow u_N - u_1 = e \left(\frac{\frac{1}{64} \left(1 - \left(\frac{1}{16} \right)^{N-1} \right)}{1 - \left(\frac{1}{16} \right)} \right) \\
&\Rightarrow u_N - u_1 = e \left(\frac{1}{60} \left(1 - \left(\frac{1}{16} \right)^{N-1} \right) \right) \\
&\Rightarrow u_N = e \left(\frac{1}{60} \left(1 - \left(\frac{1}{16} \right)^{N-1} \right) \right) + \frac{e}{30} \\
&\Rightarrow u_N = \frac{e}{20} - \frac{e}{60} \left(\frac{1}{16} \right)^{N-1} \\
&\Rightarrow u_N = \frac{e}{20} \left[1 - \frac{1}{3} \left(\frac{1}{16} \right)^{N-1} \right]
\end{aligned}$$

Additional note:

The above proof shows this is true for $N \geq 2$. When $N = 1$, $\frac{e}{20} \left[1 - \frac{1}{3} \left(\frac{1}{16} \right)^0 \right] = \frac{e}{30} = u_1$. This step completes the proof, though it is not expected in the mark scheme.

$$(i) \quad \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \left(\frac{e}{20} \left[1 - \frac{1}{3} \left(\frac{1}{16} \right)^{n-1} \right] \right) = \frac{e}{20} \lim_{n \rightarrow \infty} \left(1 - \frac{1}{3} \left(\frac{1}{16} \right)^{n-1} \right) = \frac{e}{20}$$

Therefore, u_n is a convergent sequence.

$$(ii) \quad \lim_{N \rightarrow \infty} \sum_{n=1}^N u_n = \sum_{n=1}^{\infty} \frac{e}{20} \left[1 - \frac{1}{3} \left(\frac{1}{16} \right)^{n-1} \right] = \frac{e}{20} \left(\sum_{n=1}^{\infty} 1 - \sum_{n=1}^{\infty} \frac{1}{3} \left(\frac{1}{16} \right)^{n-1} \right)$$

Since $\sum_{n=1}^{\infty} 1$ is divergent, therefore, $\sum_{n=1}^N u_n$ is a divergent series.

5. Let $f(x) = \frac{x-5}{(x+1)(3x+2)} = \frac{A}{x+1} + \frac{B}{3x+2} \Rightarrow x-5 = A(3x+2) + B(x+1)$

Sub $x = -\frac{2}{3}$, $-\frac{2}{3} - 5 = B\left(\frac{1}{3}\right) \Rightarrow B = -17$

Sub $x = -1$, $-1 - 5 = A(-1) \Rightarrow A = 6$

$\therefore f(x) = \frac{6}{x+1} - \frac{17}{3x+2}$

$f(x) = \frac{6}{x+1} - \frac{17}{3x+2}$

$= 6(1+x)^{-1} - 17(3x+2)^{-1}$

$= 6(1+x)^{-1} - 17\left[2\left(1+\frac{3x}{2}\right)\right]^{-1}$

$= 6(1+x)^{-1} - \frac{17}{2}\left(1+\frac{3x}{2}\right)^{-1}$

$= 6\left(1-x+x^2-x^3+\dots\right) - \frac{17}{2}\left(1+(-1)\left(\frac{3x}{2}\right) + \frac{(-1)(-2)}{2!}\left(\frac{3x}{2}\right)^2 + \frac{(-1)(-2)(-3)}{3!}\left(\frac{3x}{2}\right)^3 + \dots\right)$

$= 6\left(1-x+x^2-x^3+\dots\right) - \frac{17}{2}\left(1-\frac{3x}{2} + \frac{9x^2}{4} - \frac{27x^3}{8} + \dots\right)$

$= -\frac{5}{2} + \frac{27}{4}x - \frac{105}{8}x^2 + \frac{363}{16}x^3 + \dots$

Expansion of $(1+x)^{-1}$ is valid for $-1 < x < 1$

Expansion of $\left(1+\frac{3x}{2}\right)^{-1}$ is valid for $-\frac{2}{3} < x < \frac{2}{3}$

Therefore, the expansion of $f(x)$ is valid for $-\frac{2}{3} < x < \frac{2}{3}$.

Coefficient of $x^n = (-1)^n 6 + (-1)^{n+1} \frac{17}{2} \left(\frac{3}{2}\right)^n = (-1)^n \left[6 - \frac{17}{2} \left(\frac{3}{2}\right)^n\right]$

6. Let a and d be the 1st term and common difference respectively.
Sum of all even-numbered terms = 408480

$\Rightarrow \frac{444}{2} \{2(a+d) + (444-1)2d\} = 408480$

$\left[\begin{array}{l} \text{Alternatively, using } S_n = \frac{n}{2} \{1\text{st term} + \text{last term}\} \\ \Rightarrow \frac{444}{2} \{(a+d) + a + (888-1)d\} = 408480 \end{array} \right]$

$\Rightarrow 2a + 888d = 1840$

$\Rightarrow a + 444d = 920 \text{-----}(1)$

$$\begin{aligned}
\frac{a+8d}{a} &= \frac{a+20d}{a+8d} \\
\Rightarrow (a+8d)^2 &= a(a+20d) \\
\Rightarrow a^2 + 16ad + 64d^2 &= a^2 + 20ad \\
\Rightarrow 4ad &= 64d^2 \\
\Rightarrow a &= 16d \text{ ----- (2)}
\end{aligned}$$

$$\begin{aligned}
\text{Sub. (2) into (1),} \quad (16d) + 444d &= 920 \\
\Rightarrow 460d &= 920 \\
\Rightarrow d &= 2 \\
\Rightarrow a &= 16 \times 2 = 32
\end{aligned}$$

7. (i) $\ln(1+y) = \tan^{-1} x$

$$\frac{1}{1+y} \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$(1+x^2) \frac{dy}{dx} = 1+y \quad (\text{Shown})$$

(ii) $(1+x^2) \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} = \frac{dy}{dx}$

$$(1+x^2) \frac{d^3 y}{dx^3} + 2x \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 2x \frac{d^2 y}{dx^2} = \frac{d^2 y}{dx^2}$$

$$\text{When } x=0, \quad y = e^{\tan^{-1} x} - 1 = 0, \quad \frac{dy}{dx} = 1, \quad \frac{d^2 y}{dx^2} = 1, \quad \frac{d^3 y}{dx^3} = -1$$

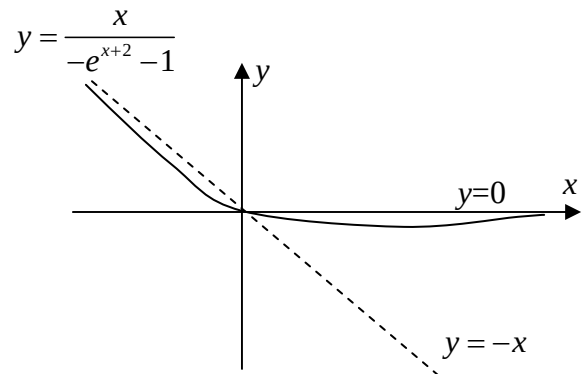
$$\therefore y = x + \frac{x^2}{2} - \frac{x^3}{6} + \dots$$

(iii) $\int_0^{\frac{1}{2}} (e^{\tan^{-1} x} - 1) dx \approx \int_0^{\frac{1}{2}} \left(x + \frac{x^2}{2} - \frac{x^3}{6} \right) dx \approx 0.14323 \approx 0.143 \text{ (to 3sf)}$

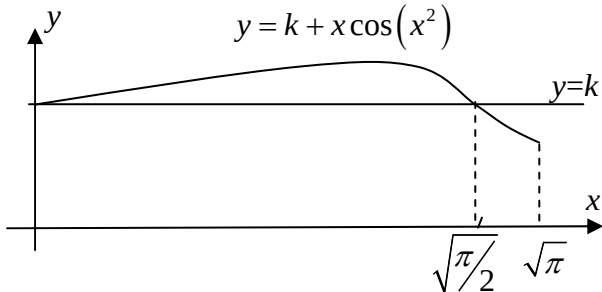
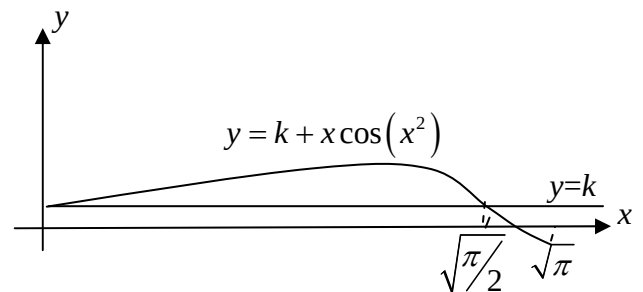
$$\begin{aligned}
\therefore \text{percentage error} &= \frac{\left| \int_0^{\frac{1}{2}} (e^{\tan^{-1} x} - 1) dx - \int_0^{\frac{1}{2}} \left(x + \frac{x^2}{2} - \frac{x^3}{6} \right) dx \right|}{\int_0^{\frac{1}{2}} (e^{\tan^{-1} x} - 1) dx} \times 100\% \\
&\approx 1.0723\% \approx 1.07\% \text{ (3sig figs)}
\end{aligned}$$

8. (i) $zy = x \Rightarrow z \frac{dy}{dx} + y \frac{dz}{dx} = 1$
 $\therefore x \frac{dy}{dx} = y(1-x-y) \Rightarrow zy \frac{dy}{dx} = y(1-x-y)$
 $\Rightarrow 1 - y \frac{dz}{dx} = 1 - x - \frac{x}{z}$
 $\Rightarrow -\frac{x}{z} \frac{dz}{dx} = -x - \frac{x}{z}$
 $\Rightarrow \frac{dz}{dx} = z + 1$
 $\Rightarrow \int \frac{1}{z+1} dz = \int 1 dx$
 $\Rightarrow \ln|z+1| = x + c, \quad \text{where } c \text{ is a constant.}$
 $\Rightarrow z+1 = \pm e^c e^x$
 $\Rightarrow z = -1 + Ae^x, \quad \text{where } A = \pm e^c (\neq 0).$
 $\therefore \frac{x}{y} = -1 + Ae^x \Rightarrow y = \frac{x}{Ae^x - 1}$

(ii) Since $y = 1$ when $x = -2$,
 $y = \frac{x}{Ae^x - 1} \Rightarrow 1 = \frac{-2}{Ae^{-2} - 1}$
 $\Rightarrow Ae^{-2} - 1 = -2$
 $\Rightarrow A = -e^2$
 $\therefore y = \frac{x}{-e^{x+2} - 1}$



9.

If $k < \sqrt{\pi}$ ORIf $0 < k < \sqrt{\pi}$ (i) Vertical translation of $-k$ units in the direction of y -axis.

$$\begin{aligned}
 \text{Exact area of } R &= \left| \int_{\sqrt{\pi/2}}^{\sqrt{\pi}} x \cos(x^2) dx \right| \\
 &= \left| \left[\frac{1}{2} \sin(x^2) \right]_{\sqrt{\pi/2}}^{\sqrt{\pi}} \right| \\
 &= \frac{1}{2} \left| \sin(\pi) - \sin\left(\frac{\pi}{2}\right) \right| \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) Required volume} &= \pi \int_{\sqrt{\pi/2}}^{\sqrt{\pi}} \left(x \cos(x^2) \right)^2 dx \\
 &\approx 2.0137 \\
 &\approx 2.01 \text{ (3sig figs.)}
 \end{aligned}$$

10. (i) Point of intersection is $\begin{pmatrix} 1 \\ 5 \\ -4 \end{pmatrix}$.

Observe that l_1 is parallel to Π_1 from their direction vectors.

Hence, l_1 lies on the plane Π_1 .

(ii) Vector parallel to $\Pi_2 = \begin{pmatrix} 3 \\ 6 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \\ -4 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$

$$\mathbf{n}_2 = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} \times \begin{pmatrix} 3 \\ 2 \\ 7 \end{pmatrix} = \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix}$$

$$\mathbf{r} \cdot \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix} = -2$$

Thus, vector equation of Π_2 is $\mathbf{r} \cdot \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix} = -2$.

(iii) $\mathbf{n}_1 = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 3 \\ 2 \\ 7 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$

$$\mathbf{r} \cdot \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} = -15$$

$$\Pi_1 : -x - 2y + z = -15 \Rightarrow x + 2y - z = 15$$

$$\Pi_2 : -3x + y + z = -2$$

$$\Pi_3 : x + 2y - z = 0$$

From the equations of Π_1 and Π_3 , observe that they have the same normal vector but the RHS are not equal. Therefore, the 2 planes are parallel and do not intersect. Thus, the 3 planes have no common point of intersection.

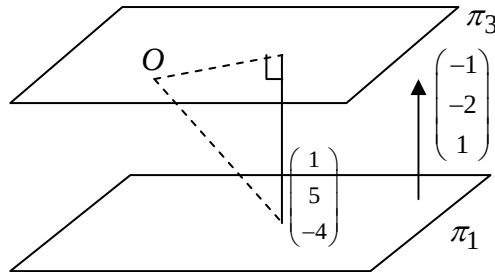
Alternatively, using GC method:

$$\text{Augmented matrix: } \begin{pmatrix} 1 & 2 & -1 & 15 \\ -3 & 1 & 1 & -2 \\ 1 & 2 & -1 & 0 \end{pmatrix}$$

$$\text{Using rref, } \begin{pmatrix} 1 & 0 & -0.42857 & 0 \\ 0 & 1 & -0.28571 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Since one of the equations (3rd equation) is invalid, the 3 planes have no point of intersection.

- (iv) From the equation of Π_3 , observe that Π_3 passes through the origin.



$$\therefore \text{Perpendicular distance} = \left| \begin{pmatrix} 1 \\ 5 \\ -4 \end{pmatrix} \cdot \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} \right| = \left| \frac{-15}{\sqrt{6}} \right| = \frac{15}{\sqrt{6}} = \frac{5\sqrt{6}}{2}$$

Alternatively:

$$\text{From the equation of } \Pi_3, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \text{ (or any other valid points) lies on } \Pi_3.$$

$$\begin{aligned}
 \therefore \text{Perpendicular distance} &= \left| \left(\begin{pmatrix} 1 \\ 5 \\ -4 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right) \cdot \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} \right| \\
 &= \left| \begin{pmatrix} 1 \\ 4 \\ -6 \end{pmatrix} \cdot \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} \right| = \left| \frac{-15}{\sqrt{6}} \right| = \frac{15}{\sqrt{6}} = \frac{5\sqrt{6}}{2}
 \end{aligned}$$

11. $\frac{dx}{d\theta} = \frac{1}{2} \sec\left(\frac{1}{2}\theta\right) \tan\left(\frac{1}{2}\theta\right),$

$$\frac{dy}{d\theta} = \left(\frac{1}{\sec\left(\frac{1}{2}\theta\right)} \right) \frac{1}{2} \sec\left(\frac{1}{2}\theta\right) \tan\left(\frac{1}{2}\theta\right) = \frac{1}{2} \tan\left(\frac{1}{2}\theta\right) \text{ (chain rule)}$$

(OR alternatively by using formula $\frac{1}{2} \tan\left(\frac{1}{2}\theta\right)$ in MF15)

$$\therefore \frac{dy}{dx} = \frac{\frac{1}{2} \tan\left(\frac{1}{2}\theta\right)}{\frac{1}{2} \sec\left(\frac{1}{2}\theta\right) \tan\left(\frac{1}{2}\theta\right)} = \cos\left(\frac{1}{2}\theta\right)$$

(a) At $\theta = \frac{\pi}{3}, \quad \frac{dy}{dx} = \frac{\sqrt{3}}{2}, \quad x = \frac{2}{\sqrt{3}}, y = \ln\left(\frac{2}{\sqrt{3}}\right)$

$$y - \ln\left(\frac{2}{\sqrt{3}}\right) = -\frac{2}{\sqrt{3}} \left(x - \frac{2}{\sqrt{3}} \right)$$

$$\Rightarrow y = -\frac{2}{\sqrt{3}}x + \ln\left(\frac{2}{\sqrt{3}}\right) + \frac{4}{3}$$

(b) $\frac{dx}{d\theta} = \frac{1}{2} \sec\left(\frac{1}{2}\theta\right) \tan\left(\frac{1}{2}\theta\right),$

$$\frac{d\theta}{dt} = \frac{d\theta}{dx} \times \frac{dx}{dt} = \frac{2 \frac{dx}{dt}}{\sec\left(\frac{1}{2}\theta\right) \tan\left(\frac{1}{2}\theta\right)}$$

$$\text{Since } \frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{d}{d\theta} \left(\frac{dy}{dx} \right) \times \frac{d\theta}{dt},$$

$$\begin{aligned}
 -0.4 &= \frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{d}{d\theta} \left(\frac{dy}{dx} \right) \times \frac{d\theta}{dt} \\
 \Rightarrow -0.4 &= \frac{d}{d\theta} \left(\cos \left(\frac{1}{2} \theta \right) \right) \times \frac{d\theta}{dt} = -\frac{1}{2} \sin \left(\frac{1}{2} \theta \right) \times \frac{d\theta}{dt} \\
 \Rightarrow -0.4 &= -\frac{1}{2} \sin \left(\frac{1}{2} \theta \right) \times \frac{2 \frac{dx}{dt}}{\sec \left(\frac{1}{2} \theta \right) \tan \left(\frac{1}{2} \theta \right)} \\
 \Rightarrow \frac{dx}{dt} &= \frac{0.4}{\cos^2 \left(\frac{1}{2} \theta \right)} = \frac{0.4}{\cos^2 \left(\frac{\pi}{4} \right)} = 0.8
 \end{aligned}$$

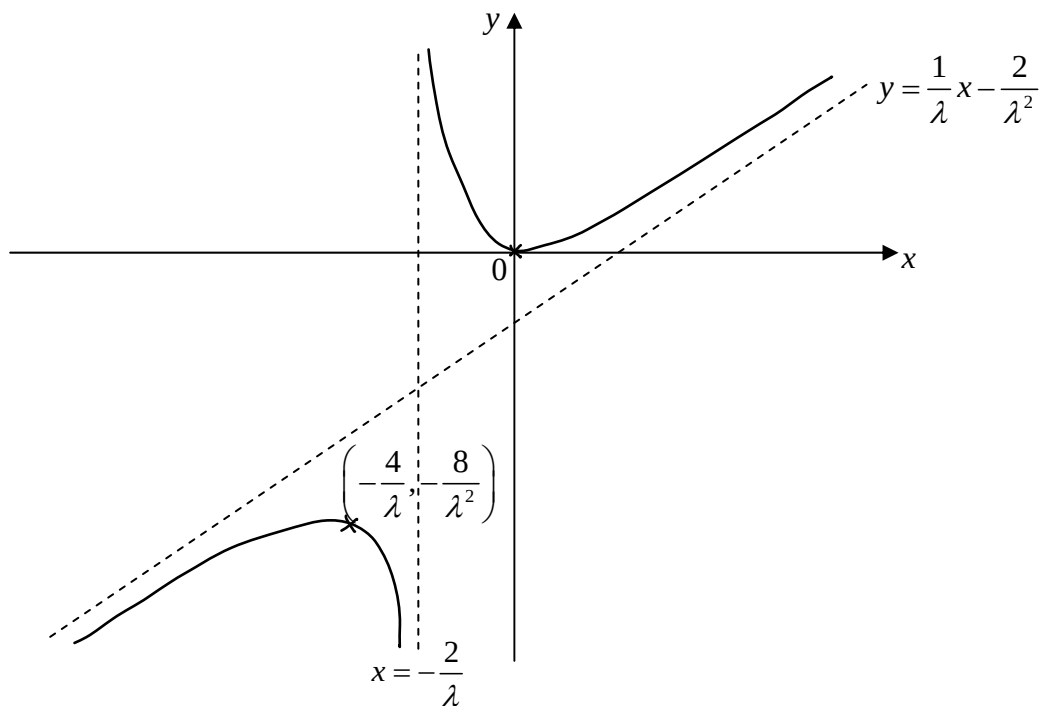
12. (i) By long division, $y = \frac{x^2}{2 + \lambda x} = \frac{1}{\lambda} x - \frac{2}{\lambda^2} + \frac{4}{\lambda^2(\lambda x + 2)}$

Oblique asymptote: $y = \frac{1}{\lambda} x - \frac{2}{\lambda^2}$, Vertical asymptote: $x = -\frac{2}{\lambda}$

(ii) $\frac{dy}{dx} = \frac{2x(2 + \lambda x) - \lambda(x^2)}{(2 + \lambda x)^2} = 0$

$\Rightarrow x = 0$ or $x = -\frac{4}{\lambda}$ (Exactly 2 points shown)

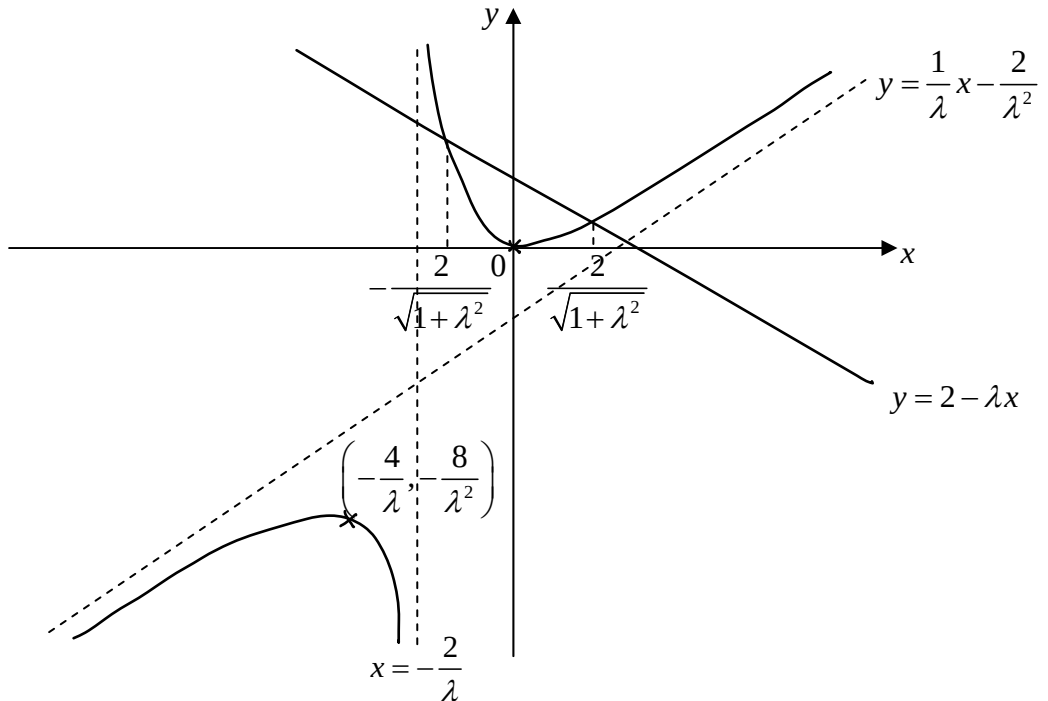
(iii)



$$\begin{aligned}
 \text{(iv)} \quad \frac{(1+\lambda^2)x^2+2\lambda x}{2+\lambda x} &\geq 2 \Rightarrow \frac{x^2+\lambda^2 x^2+2\lambda x}{2+\lambda x} \geq 2 \\
 &\Rightarrow \frac{x^2}{2+\lambda x} + \frac{\lambda^2 x^2+2\lambda x}{2+\lambda x} \geq 2 \\
 &\Rightarrow \frac{x^2}{2+\lambda x} + \frac{\lambda x(2+\lambda x)}{2+\lambda x} \geq 2 \\
 &\Rightarrow \frac{x^2}{2+\lambda x} \geq 2-\lambda x
 \end{aligned}$$

$$\begin{aligned}
 \text{When } \frac{x^2}{2+\lambda x} &= 2-\lambda x, \\
 x^2 &= (2-\lambda x)(2+\lambda x) \\
 x^2 &= 4-(\lambda x)^2 \\
 x^2(1+\lambda^2) &= 4 \\
 x &= \pm \frac{2}{\sqrt{1+\lambda^2}}
 \end{aligned}$$

Sketching the graph of $y = 2 - \lambda x$ on the same diagram,



For $\frac{x^2}{2+\lambda x} \geq 2-\lambda x$, the solution is $-\frac{2}{\lambda} < x \leq -\frac{2}{\sqrt{1+\lambda^2}}$ or $x \geq \frac{2}{\sqrt{1+\lambda^2}}$.